Scale Exercises - Section III, Level 21

The minor section starts on A.

Natural Minor One-Fret Neck Ladder: A to A

Lower the 3rd, 6th, and 7th; learn the new 3-3-2 string geometry; transpose the complete shape one fret at a time.

1. Hear the parallel-minor change

A major: A B C# D E F# G# A

A minor: A B C D E F G A

Lower degrees 3, 6, and 7 one fret.

2. Keep the 3-3-2 shape exact

Low E and A: R, R+2, R+3 (1,3,4).

D: R, R+2 (1,3).

Four-fret frame: R through R+3.

3. Preserve rhythm through the shift

Ascend 1-8; repeat root; descend 8-1.

The 16-note cell preserves the pick start.

Shift only after the final low root.

Worked example - A natural minor at low-E fret 5

Scale notes: A B C D E F G A. The two printed high roots at the center are the turnaround.

PARALLEL CHECK C# -> C | F# -> F | G# -> G

UP 1-2-b3-4-5-b6-b7-8 | DOWN 8-b7-b6-5-4-b3-2-1

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------5--7--7--5--------------------|
A  |---------5--7--8--------------8--7--5-----------|
E  |5--7--8-------------------------------8--7--5--|
```

KEY ROUTE A -> Bb -> B -> C -> C# -> D -> Eb -> E -> F -> F# -> G -> G# -> A, then reverse.

IMPORTANT Moving one fixed shape by one fret transposes it. A minor is the starting rung, not every rung.

Session protocol

1. Compare A major and A natural minor once; name the lowered 3rd, 6th, and 7th.
2. Choose 3-4 neighboring minor-key rungs for the first session.
3. Use even eighth notes and repeat the high root at the turn.
4. Pause one beat after each 16-note cell; move the whole frame one fret.
5. Repeat with a downstroke start and an upstroke start.
6. Remove the pause, connect the route, and reverse at fret 17.

Level 21 - Chromatic Natural-Minor Route

Ascending route: complete each scale cell, then move the frame +1 fret

Descending route: use the same cards in reverse order and move -1 fret

Rung	Key	Low-E root	Frame	Scale tones (degrees 1-7)
01	A natural minor	fret 5	5-8	A B C D E F G
02	Bb natural minor	fret 6	6-9	Bb C Db Eb F Gb Ab
03	B natural minor	fret 7	7-10	B C# D E F# G A
04	C natural minor	fret 8	8-11	C D Eb F G Ab Bb
05	C# natural minor	fret 9	9-12	C# D# E F# G# A B
06	D natural minor	fret 10	10-13	D E F G A Bb C
07	Eb natural minor	fret 11	11-14	Eb F Gb Ab Bb Cb Db
08	E natural minor	fret 12	12-15	E F# G A B C D
09	F natural minor	fret 13	13-16	F G Ab Bb C Db Eb
10	F# natural minor	fret 14	14-17	F# G# A B C# D E
11	G natural minor	fret 15	15-18	G A Bb C D Eb F
12	G# natural minor	fret 16	16-19	G# A# B C# D# E F#
13	A natural minor - octave	fret 17	17-20	A B C D E F G

How to use the map

Up pass:

rungs 01-13.

Turn:

finish the high-A cell at fret 17, then reverse.

Down pass:

rungs 12-01. Skip the high-A card unless starting a new round trip.

Key names:

C#, F#, and G# minor keep the spellings readable.

Goal:

hear the same minor formula as the root rises and falls.

Mastery gate before the next natural-minor level

Name every key before playing it; sing or say 1-2-b3-4-5-b6-b7; keep the four-fret frame compact; land the new root without sliding; preserve equal note length and volume through the 3-3-2 string pattern; complete the full fret-5 to fret-17 to fret-5 route with both pick starts; and recover the correct key after a deliberate random stop.

ROUND-TRIP COUNT 25 cells x 16 notes = 400 picked notes.

Level 21 - Natural Minor One-Fret Neck Ladder

Rungs 01-04: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

01 A MINOR

ROOT E5 | FRAME 5-8

Notes: A B C D E F G A

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------5--7--7--5--------------------|
A  |---------5--7--8--------------8--7--5-----------|
E  |5--7--8-------------------------------8--7--5--|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

02 BB MINOR

ROOT E6 | FRAME 6-9

Notes: Bb C Db Eb F Gb Ab Bb

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------6--8--8--6--------------------|
A  |---------6--8--9--------------9--8--6-----------|
E  |6--8--9-------------------------------9--8--6--|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

03 B MINOR

ROOT E7 | FRAME 7-10

Notes: B C# D E F# G A B

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------7--9--9--7--------------------|
A  |---------7--9--10-------------10-9--7-----------|
E  |7--9--10-------------------------------10-9--7--|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

04 C MINOR

ROOT E8 | FRAME 8-11

Notes: C D Eb F G Ab Bb C

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------8--10-10-8--------------------|
A  |---------8--10-11-------------11-10-8-----------|
E  |8--10-11-------------------------------11-10-8--|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 21 - Natural Minor One-Fret Neck Ladder

Rungs 05-08: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

05 C# MINOR

ROOT E9 | FRAME 9-12

Notes: C# D# E F# G# A B C#

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------9--11-11-9--------------------|
A  |---------9--11-12-------------12-11-9-----------|
E  |9--11-12-----------------------------12-11-9--|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

06 D MINOR

ROOT E10 | FRAME 10-13

Notes: D E F G A Bb C D

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------10-12-12-10--------------------|
A  |---------10-12-13-------------13-12-10----------|
E  |10-12-13-----------------------------13-12-10-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

07 EB MINOR

ROOT E11 | FRAME 11-14

Notes: Eb F Gb Ab Bb Cb Db Eb

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------11-13-13-11--------------------|
A  |---------11-13-14-------------14-13-11----------|
E  |11-13-14-----------------------------14-13-11-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

08 E MINOR

ROOT E12 | FRAME 12-15

Notes: E F# G A B C D E

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------12-14-14-12--------------------|
A  |---------12-14-15-------------15-14-12----------|
E  |12-14-15-----------------------------15-14-12-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 21 - Natural Minor One-Fret Neck Ladder

Rungs 09-12: play each card as one cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

09 F MINOR

ROOT E13 | FRAME 13-16

Notes: F G Ab Bb C Db Eb F

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------13-15-15-13-------------------|
A  |---------13-15-16-------------16-15-13----------|
E  |13-15-16------------------------------16-15-13-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

10 F# MINOR

ROOT E14 | FRAME 14-17

Notes: F# G# A B C# D E F#

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------14-16-16-14-------------------|
A  |---------14-16-17-------------17-16-14----------|
E  |14-16-17------------------------------17-16-14-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

11 G MINOR

ROOT E15 | FRAME 15-18

Notes: G A Bb C D Eb F G

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------15-17-17-15-------------------|
A  |---------15-17-18-------------18-17-15----------|
E  |15-17-18------------------------------18-17-15-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

12 G# MINOR

ROOT E16 | FRAME 16-19

Notes: G# A# B C# D# E F# G#

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------16-18-18-16-------------------|
A  |---------16-18-19-------------19-18-16----------|
E  |16-18-19------------------------------19-18-16-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

Level 21 - Natural Minor One-Fret Neck Ladder

Rung 13 and full-ladder consolidation

For the descending pass, read the cards in reverse order and move -1 fret

13 A MINOR (OCTAVE)

ROOT E17 | FRAME 17-20

Notes: A B C D E F G A

16-NOTE UP/DOWN CELL

```
Deg|1--2--b3-4--5--b6-b7-8--8--b7-b6-5--4--b3-2--1--|
Fng|1--3--4--1--3--4--1--3--3--1--4--3--1--4--3--1--|
e  |-------------------------------------------------|
B  |-------------------------------------------------|
G  |-------------------------------------------------|
D  |------------------17-19-19-17-------------------|
A  |---------17-19-20-------------20-19-17----------|
E  |17-19-20-----------------------------20-19-17-|
```

Finish on the low root; release pressure; move the entire frame one fret.

D-start and U-start

Full round-trip challenge

1. Start with A natural minor at fret 5.
2. Play all 13 cells without restarting the pick cycle.
3. Reverse at fret 17 and return through rungs 12-01.
4. Name each key and say the root before its first note.
5. Repeat with the opposite starting stroke.

Clean-round standard: no open strings, no dragged shift noise, no missing accidental, no hesitation at the 3-3-2 string crossings, and no tempo change at the one-fret relocations.

400 picked notes per complete round trip.

After the basic ladder is clean

Parallel contrast: play one A-major cell, then one A-natural-minor cell; hear and locate b3, b6, and b7.

Accent rotation: accent degrees 1, then b3, then 5, then b7 without changing the fingering.

Rhythm: eighth notes, straight sixteenths, then four-note bursts with one beat of rest between keys.

Direction: begin some sessions at fret 17 and descend first.

Random entry: choose any root fret from 5-17, name the minor key, play its cell, then continue in either direction.

Practice log

Date	BPM	Start stroke	Clean route?	Notes

Advance after two separate clean days, not one unusually good run.